

The Interest Split and Credit Score Timing

PrimeLendRow — Week 3 progress report. Stellar Testnet.

1. What this is about

Every loan payment carries interest, and interest is where a lending platform can quietly favour itself. A rounding choice here, an undisclosed fee there, and over thousands of payments the numbers drift in the operator's favour without anyone seeing it.

This week the interest on every repayment is divided by **one published rule**, in whole centavos, and every division is recorded. Anyone can take a payment, apply the rule, and get exactly the same answer we did — to the centavo.

The same week changes **when a credit score moves**. A score is only worth pricing a loan against if it cannot be pumped up by activity, so it now moves on outcomes — a loan repaid in full, or a default — and never in between.

2. Words you will see

Term	What it means
Interest	What the borrower pays on top of the amount borrowed.
Split	How one payment's interest is divided between the people and funds it belongs to.
Risk band	The 30% slice of interest set aside for risk. Guarantors are paid from it; the rest goes to the recovery fund.
Recovery fund	A fund built from the risk band that absorbs losses when a loan defaults.
Lending reserve	A fund built from a fixed 20% of interest that keeps the pool able to lend.
Tier	A band of credit scores. A guarantor's tier sets their share of interest.
Credit score	A number from 50 to 150 built from a member's repayment record. It sets limits and rates.
Centavo	One hundredth of a peso. All amounts are whole centavos — never decimals.
Policy	The rulebook the platform runs on. Every percentage in this report lives there, not in the code.

3. The published rule

Every peso of interest is divided the same way, every time:

Share	Goes to	Fixed?
10%	Platform fee	Fixed
20%	Lending reserve	Fixed
40%	Depositors, shared in proportion to what each has in the pool	Fixed
30%	Risk band — guarantors first, by tier, then the recovery fund	Split inside the band

The three fixed shares never move. A guarantor is paid only out of the risk band, so a guarantor's reward can never reduce what depositors earn or what the platform keeps.

Worked example

A ₱1,000 loan at 2% a month. The first payment's interest is ₱20.00:

Recipient	No guarantor	Guarantor @ 10%	Guarantor @ 25%
Guarantor — from the risk band	₱0.00	₱2.00	₱5.00
Platform fee — fixed 10%	₱2.00	₱2.00	₱2.00
Lending reserve — fixed 20%	₱4.00	₱4.00	₱4.00
Recovery fund — rest of the band	₱6.00	₱4.00	₱1.00
Depositors — fixed 40%	₱8.00	₱8.00	₱8.00
Total — always the full interest	₱20.00	₱20.00	₱20.00

The engine reproduces this table exactly, in an automated test that runs on every change.

Rounding never creates or loses a centavo

Most interest amounts do not divide evenly. On ₱0.33 of interest, 10% is 3.3 centavos — and a centavo cannot be split. So the engine follows one fixed rule:

- All amounts are whole centavos. There are no decimals anywhere in the calculation.
- The platform, reserve and depositor shares are each rounded once, to the nearest centavo. An exact half rounds to the even centavo (banker's rounding), so over many payments no side is favoured.
- The risk band is not rounded. It is whatever is left of the interest after the three fixed shares. That is what makes the parts always add back to the full interest.

The same payment always produces the same split. There is no randomness, and nothing depends on the time of day or the order payments arrive in.

The payment fee does not come out of the interest

PayPal and Stripe keep a fee on every payment. The borrower pays that fee on top of the installment, so the loan receives exactly what is due and the split is worked out on the full interest — not on whatever is left after the payment provider takes its cut.

The percentages are policy, not constants

The shares and tiers are stored in the platform's rulebook, not written into the program. When the rulebook loads, the engine checks it and refuses to start with a bad one: the four shares must add up to exactly 100%, the guarantor cap must fit inside the risk band, and the tiers must cover every score with no gaps. A broken policy fails loudly instead of splitting money wrongly. Every recorded split notes which version of the rulebook it used.

4. How much a guarantor earns

A guarantor's share of the interest depends on their **own** credit score — not the borrower's. Reputation belongs to the guarantor, so a guarantor with a strong record earns more for backing the same loan.

Guarantor's score	Share of interest	Paid on a ₱20.00 payment	Recovery fund keeps
50–79 (entry)	10%	₱2.00	₱4.00
80–104	15%	₱3.00	₱3.00
105–129	20%	₱4.00	₱2.00
130–150 (hard cap)	25%	₱5.00	₱1.00

The score is read **when each payment arrives**. A guarantor who gains standing earns more on the next payment; one who loses standing drops a tier. A guarantor below 50 earns nothing, and their share stays in the recovery fund.

The cap is 25% because the rest of the risk band is what funds the recovery fund. Paying guarantors more would leave nothing to absorb defaults.

Two guarantors at different tiers

A loan can have up to two guarantors, and they are often at different tiers. Each earns their own tier's rate on their own part of the pledge. On the ₱20.00 payment, with equal pledges from a guarantor at 10% and one at 25%:

Who	Tier	Earns
Guarantor 1	10%	₱1.00
Guarantor 2	25%	₱2.50
Both together	17.5% combined	₱3.50
Recovery fund	rest of the band	₱2.50

The guarantors' combined share is rounded once and then divided between them, so two guarantors never cost an extra centavo between them.

5. How depositors are paid

The depositors' 40% is shared between **every** member with money in the pool, in proportion to their whole balance — including money that is currently lent out or pledged. Whoever keeps savings in the pool earns on them, not only whoever's savings funded this particular loan.

Each member's share arrives as its own withdrawable amount on every repayment, and is labelled **Interest earned** in their list of deposits. On a small loan in a large pool these can be a few centavos to a few pesos — each one shown separately, so a member can see exactly where their earnings came from.

If nobody holds a balance at all, there is nobody to pay, and the depositors' share goes to the lending reserve instead of disappearing.

6. The recovery fund now absorbs defaults

Week 2 described the recovery order ending at the platform's reserve. With the recovery fund now being built from every payment, the order gains a step. It still runs outward from the borrower, and a guarantor is still only reached after the borrower has been exhausted:

Order	Whose money	Notes
1	The borrower's deposit	Their loan backing first, then anything else of theirs
2	The borrower's XLM	Seized on-chain, as shown in the Week 1 report
3	The guarantors' pledges	Only what is still unpaid, never more than pledged
4	The recovery fund	New — up to what the fund holds
5	The lending reserve	Only if the recovery fund runs out

This is what the risk band is for. The pot that is funded by interest to cover defaults is now the pot that actually covers them, and the lending reserve stays available for lending. Ordinary savers are still never touched at any step.

If a defaulted loan is later settled, the money goes back in the same order it was taken, so whichever pot absorbed a peso is the pot that gets it back.

7. When a credit score moves

A score that rises for activity can be farmed: take many tiny loans, repay each at once, and watch the number climb. So a score now moves only on outcomes:

What happens	Score change
A loan is repaid in full	+5
A loan defaults	-25
A defaulted loan is later settled, and an administrator accepts it	+25 back
A single installment is paid	No change
A deposit is made	No change
Anything else — logging in, applying, being invited	No change

Every change is written down with the score before and after, the reason, and the loan it belongs to, so any member's score can be explained from its own history.

A score change never rewrites a loan already running. The rate and limits are fixed when the loan is issued; a new score only affects the next loan. The one thing a score does change mid-loan is a guarantor's share of the interest, as described in section 4.

8. How every split is recorded

Each repayment's split is saved in the same database transaction that records the payment — they are written together or not at all. Two records are kept:

- **The split itself** — the interest and its five parts, and the version of the rulebook used.
- **Every member who received a slice** — how much, and why: a depositor's balance against the pool's, or a guarantor's pledge and tier against the loan's pledges.

Administrators see every payment's split, with every recipient, in the Interest history. Anyone else can see each repayment's split on the public Records page, with no login and no personal data — a loan is known by its reference, a guarantor as “Guarantor 1” or “Guarantor 2”.

9. What we demonstrated this week

The split engine ships with a test script that produces the evidence as a file rather than a screenshot. Its latest run:

Check	Result
The worked example, at no guarantor, 10% and 25%	Matches to the centavo
Ten amounts that do not divide evenly, from ₱0.01 to ₱12,345.67	Every one sums back exactly
The same ₱12,345.67 payment split 1,000 times	Identical every time
100,000 random repayments, with up to 40 depositors and 2 guarantors each	All sum back exactly; none negative; guarantors paid only from the band
Ten repayments with two guarantors at different tiers (below)	Every split exact
Total splits checked	100,023 — no centavo created or lost

Ten repayments across two guarantor tiers

A ₱5,000 loan over ten months, with interest falling from ₱300 to ₱30 as it is paid down. Guarantor A has a score of 70 — still in the entry tier, which runs to 79 — and earns 10%. Guarantor B has a score of 90 and earns 15%. They pledged equal amounts.

#	Interest	Guarantor A (score 70, 10%)	Guarantor B (score 90, 15%)	Recovery fund
1	₱300.00	₱15.00	₱22.50	₱52.50
2	₱270.00	₱13.50	₱20.25	₱47.25
3	₱240.00	₱12.00	₱18.00	₱42.00
4	₱210.00	₱10.50	₱15.75	₱36.75
5	₱180.00	₱9.00	₱13.50	₱31.50
6	₱150.00	₱7.50	₱11.25	₱26.25
7	₱120.00	₱6.00	₱9.00	₱21.00
8	₱90.00	₱4.50	₱6.75	₱15.75
9	₱60.00	₱3.00	₱4.50	₱10.50
10	₱30.00	₱1.50	₱2.25	₱5.25
Total	₱1,650.00	₱82.50 (5%)	₱123.75 (7.5%)	₱288.75 (17.5%)

A score of 79 was also tested and still earns 10%; a score of 80 moves up to 15%.

What we found and fixed

Testing repayments on a guarantor loan turned up a fault: the step that reads each guarantor's score failed, so a repayment on any loan with a guarantor stopped after the payment was taken. It is fixed. The same testing strengthened how an interrupted payment is handled: if a payment is taken but the platform is interrupted before recording it, the platform now asks the payment provider what happened and finishes applying the payment to the loan, rather than refunding it and asking the borrower to pay again.

10. How to check it yourself

The rule is short enough to check by hand. Take any repayment on the public Records page, multiply its interest by 10%, 20% and 40%, round each to the nearest centavo, and the three numbers will match. What remains is the risk band; the guarantor's share and the recovery fund will add up to it exactly.

To re-run the evidence yourself from the public repository, in the engine folder:

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./scripts/interest_split_evidence.sh
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It prints the report above, saves it to a file, and fails if any split does not balance.

11. What comes next

- Make a score rise wait for the loan's scheduled end, not only its last payment, so a loan paid off early cannot be repeated to build score faster.
- Lower a guarantor's score when their pledge is claimed on a default. The recovery order already takes the pledge; the score change is not yet made. A guarantor who loses score can drop a tier, so backing a bad borrower also lowers their share of every payment after.
- Show every score change on the public Records page — the score before and after, set against the loan's own dates — so anyone can see a score rise only at a completed term and never mid-loan.
- Price at least one loan above the entry tier, to show a higher score earning a better rate.